

This is part of [a process to draft and review ANZSDPs](#) by scientific experts, managed through a set of standardised procedures.

1. Background

The Animal Health Committee (AHC) National Laboratory Task Group, with administrative support from the Department of Agriculture, Fisheries and Forestry, seeks to sustain and improve the quality of livestock and livestock products and to assure market access through the application of best practice to veterinary laboratory services in Australia and New Zealand. Publication of [standard laboratory procedures](#) is an integral component of best practice.

The primary objective of Australian and New Zealand standard diagnostic procedures (ANZSDPs) is to standardise test procedures so as to ensure consistency between laboratories using methods selected for their optimal accuracy, sensitivity, specificity and robustness. Other objectives include the provision of standard methods that can be used in external proficiency testing programs and for the development of documentation for quality systems.

2. Purpose

This instruction is a guide for authors. It details the content and format for writing an ANZSDP. Further guidelines on the procedure and policy for the establishment of ANZSDPs are maintained on the Department of Agriculture, Fisheries and Forestry website—see [Editorial and approval process for ANZSDPs](#).

3. Scope

This procedure applies to the authorship of documents describing standardised laboratory testing for diseases in aquatic and terrestrial animal species in Australia and New Zealand. All AHC-approved ANZSDPs are made available in a retrievable electronic format on the department's website—see [Australian and New Zealand standard diagnostic procedures](#) (ANZSDPs).

4. Procedure

Authors should prepare procedures according to these guidelines.

4.1 Content

Each ANZSDP has 3 parts:

- Part 1—Diagnostic overview
- Part 2—Test methods
- Part 3—Reagents and kits.

4.2 Format

New authors are encouraged to familiarise themselves with the format by reading one or more existing ANZSDPs.

Unless otherwise specified, all text is formatted in Times New Roman font with 12 point size. Only 1 space and not 2 spaces is used between sentences. All pages are justified left. Every 5th line is numbered to facilitate editing and review. A short running title in bold is set in a header at the top outer margin of each page. A footer with 'ANZSDP, month, year' closest to the left margin and 'page number' at the centre, is placed on every page. For example, main headings to each section of text are bold and centred while those to sub-sections are in italics and justified to the left. See Section 6 for details of headings.

4.3 Part 1—Diagnostic overview

4.3.1 Background

The 'Diagnostic overview' is to provide general information on the aetiology, clinical signs, epidemiology, occurrence and distribution, pathology, and biosafety and biosecurity requirements of the disease in Australia and New Zealand together with the types of tests available. In this part of the ANZSDP, the description of a test does not provide technical details of the method but must indicate the type of test, its application and reliability, appropriate samples for testing, and its advantages or limitations. Key references for further information must be listed.

4.3.2 Title

The aetiological name for the disease should be used, unless it is likely to cause confusion. For example, 'bluetongue' is preferred to 'orbiviral infection' to distinguish it from non-bluetongue orbiviral infections.

4.3.3 Authors

Include authors' initials and surnames, and addresses at the time of writing.

4.3.4 Summary

The summary should be similar to that in the OIE Manual but covering only the aspects listed in Part 1.

4.3.5 Main text (do not use this as a heading)

There are a number of sections that should contain information reflecting the situation in Australia and New Zealand. These should be described under the headings listed in this section.

4.3.5.1 Aetiology

This describes the causative agent and any variations in antigenicity or pathogenicity of strains that are significant for diagnosis or epidemiology. Essential content will determine the extent of the description required.

4.3.5.2 Clinical signs

A comprehensive account of the clinical signs is not required; only the type of disease and key presenting signs are needed.

4.3.5.3 Epidemiology

This will focus on the classical aspects of how, where, when and why, including host range and reservoirs. Methods of transmission, incubation period and the agent's strategies for survival or maintenance are important. This is the place to indicate the interpretation of positive immunological/serological tests, that is, whether they indicate an animal that has been exposed to, but is now free of, the agent and is resistant to reinfection, or that the animal is latently or persistently infected with the agent.

4.3.5.4 Occurrence and distribution

This will describe the current status of the disease in animals in Australia and New Zealand. In particular, it should describe the incidence, geographical distribution, a very brief statement about any formal disease control programs that may be in place and any other features that are significant or unique to Australia and/or New Zealand. If necessary, the laboratory task group may be able to assist the author(s) to locate relevant information.

4.3.5.5 Pathology

This section should contain sufficient information to indicate the principal gross and histological lesions and the pathogenesis. It should clearly indicate the organs and tissues to be examined at necropsy and the samples to be collected for laboratory examination. It may be useful to indicate where the agent is present at highest concentration at various stages of the disease. This section is particularly important when gross and/or histopathology are the principal diagnostic tests.

4.3.5.6 Diagnostic tests (general)

This section describes the range of tests available for the particular disease. It may cover histopathology (in detail where it is a standard diagnostic test), isolation of the agent, detection of antigen including immunohistochemistry, detection of nucleic acid and detection of an immune response (which may be further divided to cover the specific serological and other immunological tests). The tests should be described in general terms, along with their applications (fitness for purpose) strengths and weaknesses. Where possible, the most reliable estimates of sensitivity and specificity should be given.

This is not the place to provide the 'bench-top' details of the tests but rather to focus on their application and interpretation. The definitive diagnostic test(s), if available, must be clearly specified. Key references for further information must be listed.

4.3.5.7 Guidance on safety and biosecurity requirements

The principal purpose of biosecurity is to prevent the escape of the pathogen from the laboratory into the national animal population. In some instances there is also a risk to human health and this may demand additional procedures than would otherwise be considered necessary from purely animal health considerations. The level of physical containment and biosecurity procedures should be related to the risk group into which the pathogen has been placed, and the detailed requirements should be appropriate to the type of organism (bacterium, virus, fungus or parasite). The minimum level of safety and biosecurity practice should be described here. However, detailed precautions in addition to those specified as the minimum, should be determined according to the degree of risk.

4.4 Part 2—Test methods

4.4.1 Background

Full technical details are described in this section. The method description should have sufficient detail to allow a competent laboratory scientist to conduct the test and, when necessary, develop an in-house manual without the need to go to secondary sources of information. There should be a clear statement of those variables that significantly affect the performance of the test.

Additional tests that may be used for diagnostic purposes locally are not to be described in this section. Laboratories will document methods used for local disease diagnostic purposes in their own Quality System.

4.4.2 Content

The test method should be described under the following headings (using standard ANZSDP formatting).

4.4.3 Principle of the test

This is a very brief description of the type of test, the way it works and the purpose of the test. For example, 'c-ELISA to detect antibodies against non-structural proteins' gives some basic information about the purpose of the test but a more suitable description would be 'c-ELISA to detect antibodies against non-structural proteins of FMD virus in pigs, cattle, sheep and goats, to allow differentiation of infected and vaccinated herds for the purpose of declaring a country free from FMD'.

4.4.4 Reagents and materials

Critical reagents and materials should be specified in sufficient detail to allow them to be obtained. Supplier and manufacturer details are not required for common laboratory chemicals and media unless

a specific brand has been shown to be critical for the test. In the case of biologicals such as test antigens, control sera or complete test kits that are available commercially, a brief description (by name) should be included in this document. A full description of the test kit (by name) or other reagents, supplier and other relevant information will be supplied in Part 3—Reagents and kits.

4.4.5 Special requirements for PCR-based methods

All PCR and other nucleic acid-based tests must undergo appropriate evaluation and laboratories should follow the guidelines for the use of [nucleic acid-based technology](#). In particular the design and layout of the PCR testing facility must be such as to minimise contamination.

Because of the multiplicity of primers in use for some procedures, it may be both difficult and unnecessary to be absolutely precise in the specifications for the test. Minor variations are possible without affecting the outcome. Performance should focus on the quality and reproducibility of the outcome. Consequently ANZSDPs including PCR methods should not be unjustifiably prescriptive except in very specific circumstances, for example, Johne's disease methods. However, where PCR is used, there should be a clear description of the target segment of the genome, its nature and the rationale underlying the use of primers directed at that site.

4.4.6 Test procedure

The test is described in a step-wise manner emphasising any precautions that should be taken. If the test is based entirely on a commercially available kit, it is appropriate to refer to the manufacturer's instructions, indicating that they should be followed in full unless there has been some variation that has been routinely adopted for use in Australia or New Zealand. This deviation would have been evaluated prior to acceptance of the kit for use as an approved method.

4.4.7 Quality control aspects

Controls to be included, and criteria necessary for the test results to be valid, must be described.

4.4.8 Interpretation of results

The criteria by which a result is classified, together with justification, should be briefly described. In some circumstances, it will be necessary to reiterate the limitations of some results in the context of the test's 'fitness for purpose'. Any issues that are not included above, such as the need for confirmatory testing, or follow-up actions that should be taken, are described here.

4.4.9 References

These are cited in the style used in the [Australian Veterinary Journal \(instructions for authors\)](#). Throughout the text, the references should be identified by superscript numbers that follow the sequence of citation. References are then listed in numerical order.

4.5 Part 3—Reagents and kits

4.5.1 Kits and reagents in common use in Australia

A list of suppliers is provided in this section. It need not be exhaustive but should include all materials (complete kits and other reagents) in common use in Australia and New Zealand laboratories at the time of writing.

4.6 Approval of tests

A test that differs in principle from an established test must be evaluated by the laboratory task group and approved by the Animal Health Committee before its inclusion in an ANZSDP. Tests will be evaluated in accordance with [procedures for new test development and evaluation](#).

Procedures requiring OIE accreditation

Laboratories testing biological materials for international accreditation must use the appropriate OIE method required under subclass 20.95 Foreign Regulatory Requirements of [ISO/IEC 17025:2005](#). The appropriate OIE method(s) will need to be used where OIE compliance is required, even if the procedures differ only minimally from those contained in the relevant ANZSDP. Authors should include a note in the ANZSDP directing Australian and New Zealand laboratories to use OIE procedures in these circumstances and to summarise the key differences between the ANZSDP and OIE methods.

4.7 Definitions

Reliability: Reliability refers to the sensitivity and specificity of a test. See [Critical factors affecting the diagnostic reliability of enzyme-linked immunosorbent assay formats \(PDF\)](#), Scientific and Technical Review of the Office International des Epizooties 1998, 17(2): 550–561.

Consistency: Degree of variation between reagent lots (serials) used in an assay, e.g. reagent lots (serials), need to be evaluated for consistency so variability is not introduced into the assay as new lots are acquired. Whenever possible, it is important to change only one reagent at a time to avoid the compound problem of evaluating more than one variable at a time. (OIE Manual 5th edition 2004, p. 29).

Accuracy: Level of agreement between a test value and the expected value for the reference standard of known activity titre (OIE Manual 5th edition 2004, p. XVII).

Sensitivity:

Sensitivity (analytical)—Smallest detectable amount of analyte in question; analyte may include antibiotics, antigens, nucleic acids or live organisms.

Sensitivity (diagnostic)—Proportion of known infected reference animals that test positive in the assay; infected animals that test negative are considered to have false-negative results.

Sensitivity (relative)—Proportion of reference samples defined as positive by one or a combination of test methods that also test positive in the assay being compared (OIE Manual 5th edition 2004).

Specificity:

Specificity (analytical)—Degree to which analytes other than that in question react in an assay; the higher the level of cross-reaction, the lower the analytical sensitivity.

Specificity (diagnostic)—Proportion of known uninfected reference animals that test negative in the assay; uninfected reference animals that test positive are considered to have false-positive results.

Specificity (relative)—Proportion of reference samples, defined as negative by one or a combination of test methods, that also test negative in the assay being compared (OIE Manual 5th edition 2004).

Robustness (syn. for ruggedness): ‘The robustness of an analytical method is a measure of its capacity to remain unaffected by small, but deliberate variations in method performance parameters and provides an indication of its reliability during normal usage.’—Codex Alimentarius Commission, April, 2005. ‘A robust assay is not easily affected by physical factors, operators, or other geographical location where used or where samples come from’—The ELISA Guidebook, John R Crowther, 2001, Methods in Molecular Biology, Vol. 149, Humana Press, p. 301.

5. References

OIE 2004, [OIE Manual of Diagnostic Tests and Vaccines for Terrestrial Animals \(mammals, birds and bees\)](#), 5th edition.

Schrijver RS and Kramps JA 1998, ‘[Critical factors affecting the diagnostic reliability of enzyme-linked immunosorbent assay formats \(PDF\)](#)’, Scientific and Technical Review of the Office International des Epizooties, vol. 17, no. 2, pp. 550–561.

Crowther JR 2001, [The ELISA Guidebook, Methods in Molecular Biology Vol 149](#), Humana Press, New York City, p. 301.

The following document should be read as a supplement to this standard operating procedure:

OIE Terrestrial Manual 2013, [Principles and methods of validation of diagnostic assays for infectious diseases \(PDF\)](#), chapter 1.1.6.

6. ANZSDP manuscript layout—headings (sequence)

{Title (16 pt, bold and centred)}

{Authors (12 Pt, bold and justified left)}

{Address of authors (12 Pt, bold and justified right)}

Part 1—Diagnostic overview (12 Pt, bold and centered)

Summary (12 Pt bold and centred; text in italics)

Aetiology (12 Pt bold and centred)
Clinical signs (12 Pt bold and centred)
Epidemiology (12 Pt bold and centred)
Occurrence and distribution (12 Pt bold and centred)
Pathology (12 Pt bold and centred)
Diagnostic tests (12 Pt bold and centred)
Safety and biosecurity requirements (if appropriate, 12 Pt bold and centred)

Part 2—Test methods (12 Pt bold and centred)
{Test 1} Title (12 Pt bold and centred)
Principle of the test (12 Pt italics and justified left)
Reagents and materials (12 Pt italics and justified left) (note PCR requirements)
Test procedure (12 Pt italics and justified left) (include quality control aspects)
Interpretation of results (12 Pt italics and justified left) (include additional comments)
{Test 2} as for test 1, etc.
References (12 Pt bold and centred)

Part 3—Reagents and kits (12 Pt bold and centred)
Reagent titles (12 Pt bold and centred; in sequence of test number)
Suppliers in Australia (12 Pt bold and centred)
Suppliers in New Zealand (12 Pt bold and centred)
Appendix (if required)
Note on approved tests
Note on testing biological materials for OIE compliance

Source: <https://www.agriculture.gov.au/agriculture-land/animal/health/laboratories/procedures/anzsdg-reviews/author-guidelines>